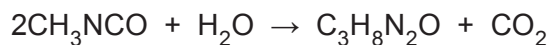


5. The release of a cloud of highly toxic methyl isocyanate (**MIC**) gas from a pesticide plant in Bhopal led to the deaths of thousands of people. On the night of the disaster **MIC** reacted with water that leaked into a storage tank, causing an explosion. As a result, **MIC** was released into the atmosphere.

The chemical formula of MIC is CH_3NCO and the equation for the reaction of MIC with water is shown below.



- (a) Calculate the relative molecular mass (M_r) of MIC using information from the Periodic Table on page 16. Show your workings, starting with the relative atomic mass (A_r) of each element found in **MIC**. [4]

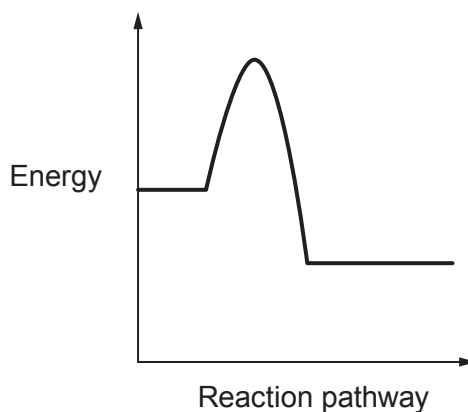
$M_r =$

- (b) Approximately 30 000 kg of MIC gas was released. Calculate the number of moles released to three significant figures using your answer in (a). [3]

number of moles =



- $$2(\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}) + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{C}-\text{N}(\text{H})-\text{C}(=\text{O})-\text{N}(\text{H})-\text{CH}_3 + \text{CO}_2$$
- MIC 1,3-dimethylurea



Use **all** the information above to explain why the reaction between MIC and water resulted in an explosion in terms of reaction pathway, bond energies and thermal runaway. [6]

[6 QER]

